1.

Operating system controls:

A. Keyboard only
B. CPU only
C. All hardware and software
D. Internet

Answer: C

2.

Which of the following is not a function of an operating system?

A. Memory management
B. Process management
C. Application designing
D. File management

Answer: C

3.

Which component of the OS is responsible for managing processes?

A. Scheduler
B. File management
C. Device driver
D. Compiler

Answer: A

4.

Which program is first executed during booting?

A. MS Word
B. POST
C. BIOS
D. Antivirus

Answer: C

5.

In the operating system magic number is used to:

A. Encrypt password
B. Identify the file format/type
C. Control memory size
D. Schedule processes

Answer: B

6.

In a multi-threaded process, threads:

A. Do not share open files
B. Share stack and registers
C. Share code and data
D. Cannot run in parallel

Answer: C

7.

Batch systems were mostly used in:

A. Supercomputers
B. Modern desktops
C. Mainframe computers
D. Mobile Devices

Answer: C

1.

Which operating system allows a user to run multiple applications at the same time?

A. Single-tasking Operating System
B Real-Time Operating System (RTOS)
C. Multi-tasking Operating System
D. Batch Operating System

Answer: C

2.

Which of the following correctly shows the order of process states?

A. New -> Running -> Waiting -> Terminated
B. New -> Ready -> Running -> Terminated
C. New -> Running -> ready -> Waiting
D. Ready -> New -> Running -> Terminated

Answer: B

3.

What is the convoy effect in the operating system?

A. When all processes are executed in parallel
B. When a low-priority process delays high-priority ones
C. When multiple short processes get stuck behind a long process
D. When all processes enter critical section at the same time

Answer: C

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4.

What is turnaround time?

A. Time taken to respond to the first request
B. Time taken for process completion
C. Time spent waiting in the ready queue
D. Time spent in I/O

Answer: B

5.

Which of the following is a data structure used by the kernel to manage processes?

A. Process Control Block (PCB)
B. Ready Queue
C. Jo b Queue
D. All of the above

Answer: D

6.

Which of the following is not typically stored in the Process Control Block (PCB)?

A. Program counter
B. Page table
C. File descriptors
D. Network address

Answer: D

7.

Which IPC mechanism is fastest in terms of data exchange between processes?

A. Message Queue
B. Pipe
C. Shared Memory
D. Semaphores

Answer: C

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1.

What is the main use of Resource Allocation Graph (RAG) ?

A. To allocate memory
B. To manage CPU time
C. To detect and avoid deadlocks
D. To prevent paging

Answer: C

2.

In deadlock prevention, how is the hold and wait condition prevented

A. Allowing processes to request all resources at once
B. Allowing circular wait
C. Not allowing mutual exclusion
D. Terminating all processes

Answer: A

3.

Which of the following IPC mechanisms allows communication between processes on different machines?

A. Pipe
B. Socket
C. Shared Memory Model
D. Message Queue

Answer: B

4.

Which of the following is a drawback of deadlock prevention?

A. Requires too much memory
B. It makes the system non-deterministic
C. Leads to low resource utilization
D. Causes compiler error

Answer: C

5.

The major difference between semaphore and mutex is:

A. Mutex can be shared between processes
B. Only one thread can acquire a mutex at a time, semaphores can allow, multiple
C. Semaphores are not used in synchronization
D. Mutex is used in message passing

Answer: B

6.

In semaphores, if a process calls wait() and the semaphore value is 0, what happens?

A. Process executes normally
B. Process is blocked until the value is positive
C. Process crashes
D. Process enters the critical section

Answer: B

7.

If S =5, and four wait(s) operations followed by two signal(s) operations executed, what is the final value of S?

A. 1
B. 2
C. 3
D. 5

Answer: C

1.

Which strategy leads to the least external fragmentation in most cases?

A. Best Fit
B. First Fit
C. Worst Fit
D. None of the above

Answer: A

2.

In paging logical address is divided into:

A. Page number and frame number
B. Page number and offset
C. Offset and segment
D. Page size and block size

Answer: B

3.

Fragmentation in Operating Systems occurs due to:

A. Improper file naming
B. Improper I/O scheduling
C. Inefficient memory allocation
D. CPU Scheduling

Answer: C

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4.

Optimal page replacement algorithm replaces the page:

A. That will not be used for the longest period of time in future
B. That is least recently used
C. That is first added
D. That has lowest priority

Answer: A

5.

Which of the following statements is true about thrashing?

A. It improves system performance
B. Occurs when CPU is underutilized
C. Occurs due to frequent page faults
D. Happens when disk I/O is disabled

Answer: C

6.

Which of the following page replacement algorithms may cause Belady's Anomaly?

A. FIFO
B. Optimal
C. Random
D. None of the above

Answer: B

7.

SSTF stands for:

A. Smallest Storage Transfer Format
B. Short Sector Transfer Function
C. Sorted Seek Track Format
D. Shortest Seek Time First

Answer: D

1.

The Operating System keeps information about files in a structure is called:

A. File control block (FCB)
B. File table
C. Access table
D. Directory

Answer: A

2.

The Operating system component that handles file creation, deletion, reading, and writing is called:

A. File server
B. File system
C. File handler
D. Data loader

Answer: B

3.

Which of the following is a type of file access method?

A. Sequential
B. Random
C. Indexed
D. All of the above

Answer: D

4.

What is a major disadvantage of the sequential access method?

A. Too fast
B. High cost
C. Cannot jump to specific record
D. Supports only binary files

Answer: C

5.

FAT32 stands for:

A. File Arrangement Type
B. File Addressing table
C. File Access Tracker
D. File Allocation Table

Answer: D

6.

Which of the following is not a function of the file system?

A. File naming
B. File protection
C. File editing
D. File access control

Answer: C

7.

The main difference between FAT32 and NTFS is:

A. NTFS supports journaling and permissions
B. FAT32 is faster
C. NTFS is only used in smartphones
D. FAT32 supports large files

Answer: A

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1.

In which of the following interrupt types does the CPU check periodically whether an interrupt has occurred?

A. Hardware interrupt
B. Software interrupt
C. Vectored interrupt
D. Polling

Answer: D

2.

What is ISR?

A. A routine that saves memory
B. A program that handles interrupts
C. A disk driver
D. A cache memory controller

Answer: B

3.

Which is a memory management technique that divides memory into equal fixed-size blocks?

A. Segmentation
B. Paging
C. Fragmentation
D. Swapping

Answer: B

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4.

The memory unit that communicates directly with the CPU is:

A. Secondary memory
B. Main memory (RAM)
C. Cache memory
D. Virtual memory

Answer: C

5.

Virtual memory is:

A. Part of main memory
B. Part of cache
C. Space on the hard disk used as RAM
D. A type of ROM

Answer: C

6.

External fragmentation is a problem in:

A. Paging
B. Fixed partitioning
C. Segmentation
D. Segmentation and contiguous memory

Answer: D

7.

What is the main function of the Memory Management Unit (MMU)?

A. Input / Output control
B. Address translation
C. Disk scheduling
D. Power management

Answer: B

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1.

What is an operating system?

A. A collection of hardware components
B. A type of application software
C. A system software that manages hardware and software
D. A programming language

Answer: C

2.

Which of the following is not a type of a computer?

A. Supercomputer
B. Smartbook
C. Mini computer
D. mainframe computer

Answer: B

3.

What was used for input in the first generation of OS?

A. keyboard
B. Punchcard
C. Touch Screen
D. Mouse

Answer: B

4.

Which of the following is not an operating system?

A. Windows
B. Linux
C. macOS
D. Oracle

Answer: D

5.

What is the first program that runs when we start the computer?

A. MS paint
B. Browser
C. Operating system
D. Calculator

Answer: C

6.

What is a magic number in the context of an executable file?

A. A randomly generated number
B. A password to protect files
C. A unique number used to identify the file format
D. A serial number for programs

Answer: C

7.

What type of file format is .exe?

A. Image file
B. Executable file
C. Document file
D. Compressed file

Answer: B

1.

In which mode does the user run applications like MS world, chrome or games?

A. Kernel mode
B. Admin mode
C. User mode
D. Debug mode

Answer: C

2.

What does POST stand for during booting?

A. Power on self test
B. Power on Software test
C. Programmed Operating system test
D. Personal OS start

Answer: A

3.

Which interface uses icons and buttons to interact with the user?

A. CUI
B. GUI
C. BIOS
D. CLI

Answer: B

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4.

What is usually the default name of the final output (executable) file after compilation in linux?

A. main.out
B. final.exe
C. program.o
D. a.out

Answer: D

5.

What is the .i file in the compilation process?

A. Compiled code
B. Preprocessed source code
C. Executable code
D. Assembly code

Answer: B

6.

Which of the following is true about Resident Monitors?

A. They are reloaded every time a job runs
B. They remain in memory throughout the execution
C. They are stored on the hard disk permanently
D. They are replaced by device drivers

Answer: B

7.

Multiprogramming improves system performance by_____.

A. Increasing I/O devices
B. Allowing parallel printing
C. Reducing idle time of CPU
D. Using graphical interface

Answer: C

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1.

Which state means the process is waiting to be assigned to a CPU?

A. New
B. Running
C. Waiting
D. Ready

Answer: D

2.

What does the OS do when a running process needs to wait for I/O?

A. Moves it to the ready state
B. Deletes it
C. Moves it to the waiting state
D. Terminates it

Answer: C

3.

Which scheduling algorithm gives the minimum waiting time for a set of processes?

A. Shortest Job first (SJF)
B. Round Robin
C. First-Come, First-Served (FCFS)
D. Priority Scheduling

Answer: A

4.

Which of these is a drawback of non-preemptive scheduling?

A. CPU switches too often
B. Context switching overhead
C. One long process can block others
D. It can not be used for batch system

Answer: C

5.

What does aging prevent starvation?

A. By decreasing burst time
B. By giving more memory
C. By gradually increasing the priority of waiting processes
D. By terminating low-priority processes

Answer: C

6.

Which scheduling algorithm selects the job with the least time remaining to execute?

A. FCFS
B. SRTF
C. SJF
D. Round Robin

Answer: B

7.

Which one is not considered a CPU scheduling criterion?

A. Context Switching
B. Waiting Time
C. Throughput
D. Turnaround Time

Answer: A

1.

A cooperative process is:

A. Runs on its own without any communication
B. Shares data and resources with other processes
C. Is terminated after execution
D. Is only used in batch system

Answer: B

2.

The main problem with multiple processes in the critical section is:

A. High memory usage
B. Slow execution
C. Data inconsistency or corruption
D. Process starvation

Answer: C

3.

Which of the following is not a necessary condition of deadlock?

A. Mutual exclusion
B. Hold and wait
C. Preemption
D. Circular wait

Answer: C

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4.

Which semaphore allows only one process in the critical section at a time?

A. Counting semaphore
B. Binary semaphore
C. Spinlock
D. None of the above

Answer: B

5.

In multiprogramming system, when one program is waiting for I/O, then CPU:

A. Wait with the program
B. Remains idle
C. Switches to another program
D. Shuts downs

Answer: C

6. Linux or

Which System call is used to create a new process in Unix ?

A. Fork()
B. Create()
C. Process_create()
D. Start

Answer: A

7.

A counting semaphore initializes to 3. Three processes call wait(), and one process calls signal(). What is the final value of the semaphore?

A. 0
B. 1
C. 2
D. 3

Answer: B

1.

MMU helps in:

A. File transfer
B. Address translation
C. Scheduling tasks
D. Disk formatting

Answer: B

2.

Internal fragmentation occurs when:

A. Memory is too small for any process
B. Memory block is larger than required
C. Memory is divided in unequal parts
D. Process size is larger than the memory block

Answer: B

3.

A computer has a memory access time of 100 nanoseconds. The time taken to handle a page fault is 5,000 nanoseconds, if the page fault rate is 0.01, what is the effective access time?

A. 144 ns
B. 140 ns
C.142 ns
D. 149 ns

Answer: D

EAT = (1-P) × mA + P × (pagefault + mA)
= (1-0.01) × 100 + 0.01 × (5000 + 100)
= 0.99 × 100 + 0.01 × 5100
= 99 + 51
= 150ns

4.

Which file system is commonly used in Windows?

A. EXT3
B. NTFS
C. HFS
D. XFS

Answer: B

5.

Which of the following is an example of a disk storage device?

A. RAM
B. CPU
C. Hard Disk Drive (HDD)
D. Monitor

Answer: C

6.

Which allocation method is fastest in practice?

A. Best Fit
B. Worst Fit
C.First Fit
D. Next Fit

Answer: C

7.

In segmented memory management, which causes a segmentation fault?

A. Address within segment limit
B. Address greater than limit
C. Address equal to base
D. Address in valid range

Answer: B

1.

Which of the following is a volatile memory?

A. ROM
B. RAM
C. Hard disk
D. Pen drive

Answer: B

2.

What does a multitasking OS allow?

A. Only one job at a time
B. Many users at once
C. Many tasks at the same time
D. None of the above

Answer: C

3.

Which type of memory is used to store frequently accessed data?

A. ROM
B. Cache
C. CD-ROM
D. Register

Answer: B

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4.

What does EEPROM stand for?

A. Electrically Erased Primary Read-Only Memory
B. Electrically Erasable Programmable Read-Only Memory
C. Externally Erasable Programmable Read-Only Memory
D. Extended Erased Programmable Read-Only Memory

Answer: B

5.

Which of the following is the fastest storage area in the CPU?

A. Register
B. RAM
C. Hard disk
D. Cache

Answer: A

6.

1 Kilobyte(KB) is equal to?

A. 1000 bytes
B. 100 bytes
C. 1024 bytes
D. 512 bytes

Answer: C

7.

Arrange the following in increasing order of memory size:

1. KB 2. GB 3. TB 4. MB

A. KB < MB < GB < TB
B. MB < KB < GB < TB
C. GB < KB < MB < TB
D. TB < GB < MB < KB

Answer: A

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